

StatLean

A Lean 4 Formalization of Mathematical Statistics

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Why Formalize Statistics?

The Problem

- ▶ In the LLM era, training on statistical proofs requires high-quality formal data
- ▶ Lean 4 + Mathlib provide a mature platform for mathematics
- ▶ But **no comprehensive statistics library** exists yet

Our Approach

- ▶ **AI-assisted formalization**: LLM-driven proof search with human guidance
- ▶ Build on **Mathlib** (measure theory, probability, analysis)
- ▶ **Automated curation**: verified lemmas auto-indexed into reusable library

Benefits

1. **Reusability**: proved lemmas compose into larger results
2. **Discovery**: formalization reveals hidden structure and missing conditions

Tech Stack

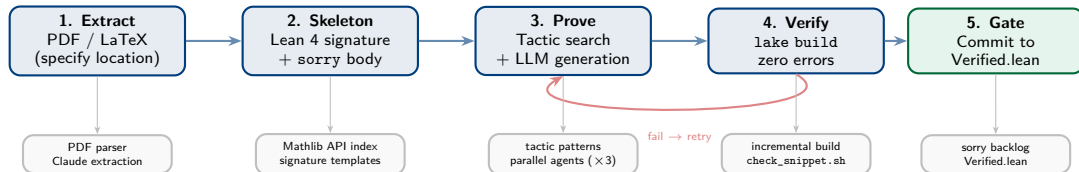
- ▶ Lean 4.28 + Mathlib (latest)
- ▶ Lake build system

Scale

45	Lean source files
14,400+	lines of Lean code
490+	formal declarations
38	fully verified modules (zero sorry)
6	remaining sorry gaps

`lake build Statlean.Verified` → zero warnings

End-to-End Formalization Pipeline



Input: PDF/LaTeX with specified theorem location,
or direct text description of the theorem to formalize

Output: machine-verified Lean 4 proof

Automation: Steps 1–4 driven by Claude Code

Human: review gate at Step 5,
mathematical guidance

Step 1–2: From Textbook to Lean Skeleton

Theorem Extraction

1. Parse PDF / LaTeX source
2. Identify theorem statement, hypotheses, notation
3. Map to Mathlib types and conventions
4. Resolve ambiguity with textbook context

Signature Design Principles

- ▶ Follow Mathlib conventions (universe-polymorphic, typeclass args)
- ▶ All conditions explicit (no “clearly integrable”)
- ▶ Prefer MemLp over ad-hoc integrability

Mathlib API Search (3-tier, mandatory)

Tier 1: Static index (650+ entries)
80% hit rate, 0 token cost

miss

Tier 2: #check / exact?
Precise but slow (~30s)

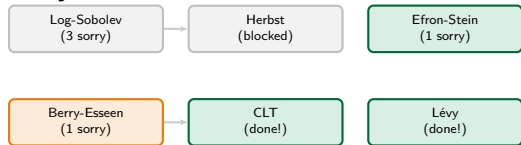
miss

Tier 3: grep Mathlib source
Last resort, targeted directories

Avoids redundant Mathlib searches — the biggest token cost in LLM-based proving.

Step 3–4: Prove and Verify

Sorry DAG Scheduler

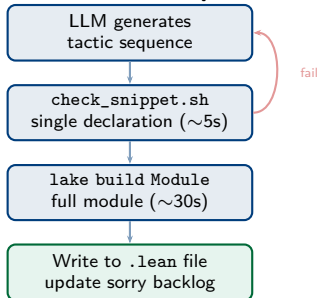


- ▶ **Leaves first:** attack goals with no downstream dependencies
- ▶ **Skip blocked:** don't waste cycles on dependency chains
- ▶ **Parallel:** up to 3 agents on independent goals

Tactic Pattern Library

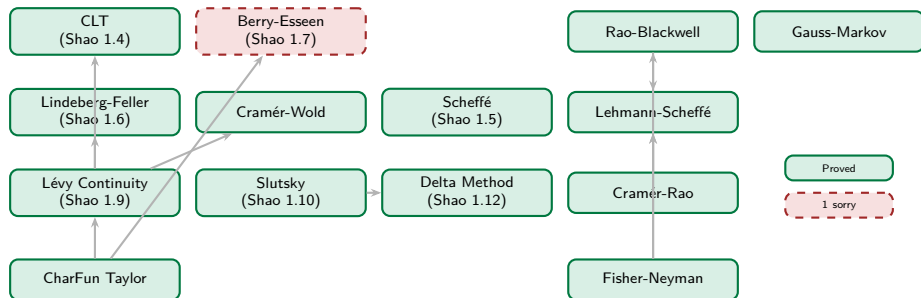
- ▶ Indexed by goal shape (e.g. " $\int f d\mu = 0$ " $\rightarrow$ `integral_condExp`)
- ▶ Only verified patterns (`lake build` passed)

Verification Loop



Key: sub-lemmas written to disk *immediately* after verification — context exhaustion won't lose progress.

Theorem Dependency Map



What Claude Code Does

Task	Method
Find Mathlib API	3-tier index search
Generate proof	Pattern library + LLM
Verify	lake build (incremental)
Manage sorry DAG	YAML backlog + priority
Parallelize	Up to 3 concurrent agents
Persist progress	Immediate disk write

What the Human Does

- ▶ Mathematical insight for hard blockers
- ▶ Decompose complex goals into sub-lemmas
- ▶ Review and approve commits
- ▶ Decide attack priority

Quantitative Results

16 $\rightarrow$ 6	sorry gaps closed
100	lines for CLT proof
690	lines for Lindeberg-Feller
260	lines for Lévy continuity

Proof Difficulty Grading and Cost

Sorry Grading System (calibrated from StatLean attack logs)

	Type	Lines	Time	Tokens	Opus \$
S	Thin wrapper	1–5	2–5 min	5–20K	0.2–0.7
A	Compose APIs	10–30	5–15 min	20–50K	0.7–1.7
B	Arithmetic	30–80	15–40 min	50–100K	1.7–3.3
C	Structural	80–200	0.5–1.5 hr	100–160K	3.3–5.3
D	Infra missing	150–400	1–3 hr	150–300K	5–10
E	Research-level	200–700+	3–10+ hr	300–500K+	10–17+

Opus 4.6 blended: \$33/M tok (70% in \$15 + 30% out \$75)

Grading Rule: grep goal in API index → exact match
S, composable **A**, algebra **B**, new lemmas **C**, API missing **D**, route unclear **E**

Validated Examples

Theorem		Tok	
Kolmogorov 0-1	S	95K	✓
Multivar. CLT	A	141K	✓
Lyapunov	B	157K	✓
Kolmogorov max	C	164K	1 sorry
Glivenko-Cantelli	C	156K	1 sorry
CLT	D	200K	✓
Lindeberg-Feller	D	300K	✓
Lévy cont.	D	250K	✓
Hypercontract.	E	400K	✗
Stieltjes inv.	E	300K	✗

S–D: \$0.2–\$10 per theorem

E: \$10+ with no success guarantee

Future I: LLM Benchmark for Theorem Proving

Goal: evaluate LLM proving on *real* statistics theorems

18 Problems (from StatLean verified theorems)

Difficulty	#	Example
Easy	5	Slutsky, Gauss-Markov, MSE
Medium	8	Rao-Blackwell, Scheffé
Hard	4	Lindeberg-Feller, USLLN
Open	1	Berry-Esseen (unsolved)

13 Models across 7 providers

Anthropic (Opus/Sonnet/Haiku) · OpenAI (GPT-5.2/o3) · Google (Gemini 3.1 Pro/2.5 Flash) · DeepSeek (V3.2 Chat/Reasoner) · Qwen 3.5 · MiniMax M2.5 · Llama 4 · **DeepSeek-Prover-V2** (proof-specialized)

Experiment Design

Condition	Content
Bare Skill	theorem + imports only + tactic patterns + API index
Budget	1-round vs. 4-round retry

Matrix: $13 \times 18 \times 2 \times 2 \times N = 936N$ runs

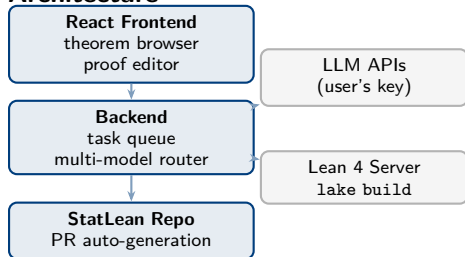
Metrics: success rate, cost/theorem, Pass@k, first-pass rate, failure taxonomy

Status: harness ready (3,500 LoC, 9 tests passing)

Future II: Web Platform for Community Contributions

Vision: lower the barrier — anyone can contribute verified proofs via web interface

Architecture



User Workflow

1. Browse open sorry goals (prioritized)
2. Select a goal, choose model + budget
3. System runs proving pipeline
4. Review result, submit as PR

Key Design Decisions

- ▶ **BYOK:** users provide own API key — zero hosting cost
- ▶ **Sandboxed:** isolated Lean environment per task
- ▶ **Reproducible:** full logs stored per run
- ▶ **Open data:** proving traces for research

GitHub:

github.com/mockingbird-gan/statlean4